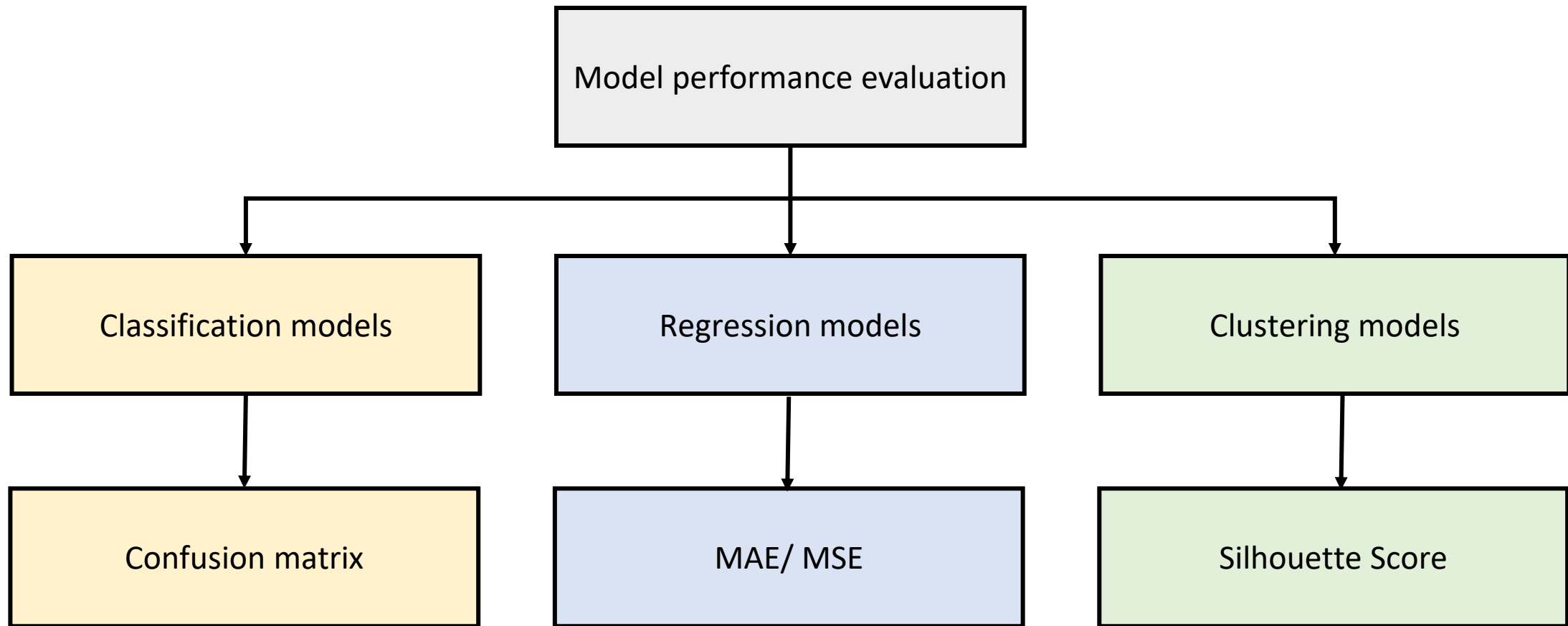


Evaluation metrics

Evaluation metrics for classification and regression

1. Accuracy, precision, recall, f1-score
2. Mean Square Error, R-square error

Evaluation of model performance



Model performance evaluation for classification models

Classification Performance Evaluation Metrics

Confusion Matrix

What is a Confusion Matrix?

- True Positive
- True Negative
- False Positive – Type 1 Error
- False Negative – Type 2 Error

Why need a Confusion matrix?

Precision vs Recall

F1-score

Introduction

A Confusion matrix is an $N \times N$ matrix used for evaluating the performance of a **classification model**, where **N** is the number of **target classes**.

The matrix compares the **actual** target values with those **predicted** by the machine learning model.

This gives us an idea about how **well our classification model is performing** and **what kinds of errors it is making**.

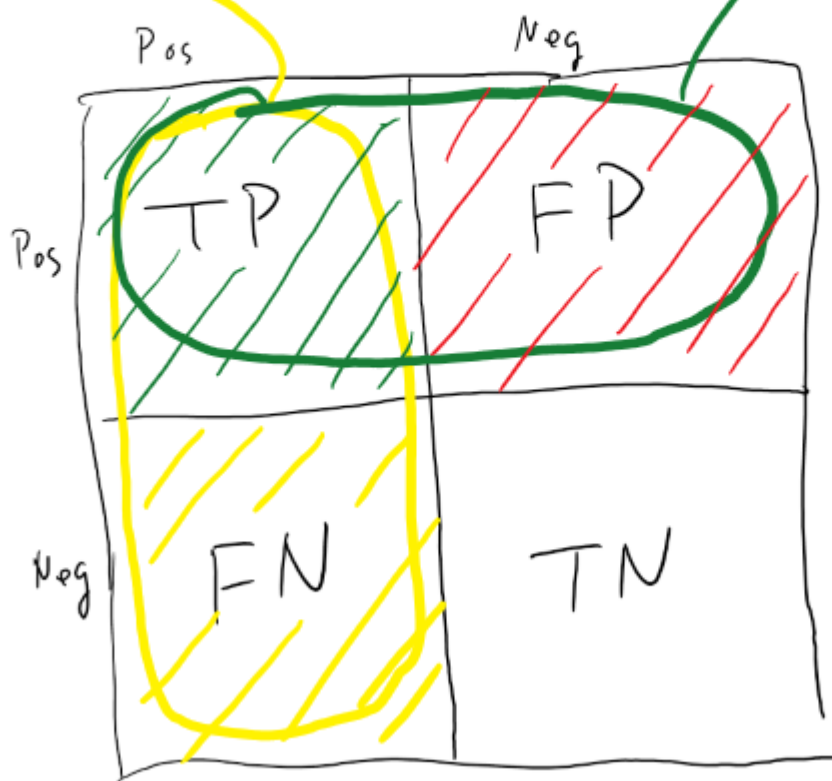
Confusion matrix

Input attributes			Target/ output		Confusion matrix
A1	A2	A3	Actual	Predicted	
-	-	-	Y	Y	True positive
-	-	-	Y	N	False negative
-	-	-	N	Y	False positive
-	-	-	N	N	True negative

$$\text{Recall} = \frac{TP}{TP + FN} \quad \text{Actual}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

Predicted



$$\text{Accuracy} = \frac{TP + TN}{\text{Total}}$$

Confusion Matrix

For a binary classification problem: 2 x 2 matrix with 4 values:

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
PREDICTED VALUES	POSITIVE	TP	FP
	NEGATIVE	FN	TN

True Positive (TP)

- The **actual value** is **positive**, and the model predicted a positive value

True Negative (TN)

- The **actual value** is **negative**, and the model predicted a negative value

False Positive (FP) – Type 1 error

- The **actual value** is **negative**, but the model predicted a positive value
- Also known as the **Type 1 error**

False Negative (FN) – Type 2 error

- The **actual value** is **positive**, but the model predicted a negative value
- Also known as the **Type 2 error**

		ACTUAL VALUES	
		POSITIVE	NEGATIVE
PREDICTED VALUES	POSITIVE	560	60
	NEGATIVE	50	330

The different values of the Confusion matrix would be as follows:

- **True Positive (TP)** = 560; meaning 560 positive class data points were correctly classified by the model
- **True Negative (TN)** = 330; meaning 330 negative class data points were correctly classified by the model
- **False Positive (FP)** = 60; meaning 60 negative class data points were incorrectly classified as belonging to the positive class by the model
- **False Negative (FN)** = 50; meaning 50 positive class data points were incorrectly classified as belonging to the negative class by the model

Metrics based on Confusion Matrix Data

Accuracy

- Accuracy is used to measure the performance of the model. It is the ratio of Total correct instances to the total instances.

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

$$Accuracy = \frac{560 + 330}{560 + 330 + 50 + 60} = 0.89$$

Metrics based on Confusion Matrix Data

Precision

- Precision is a measure of **how accurate a model's positive predictions** are. It is defined as the ratio of true positive predictions to the total number of positive predictions made by the model.

$$Precision = \frac{TP}{TP + FP}$$

$$Precision = \frac{560}{560 + 60} = 0.903$$

Of all observations predicted as positive, how many were actually positive?

Metrics based on Confusion Matrix Data

Recall

- [Recall](#) measures the effectiveness of a classification model in identifying all **relevant instances** from a dataset. It is the ratio of the number of true positive (TP) instances to the sum of true positive and false negative (FN) instances.

$$\text{Sensitivity or Recall} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{560}{560 + 50} = 0.918$$

Of all actual positive observations, how many were identified correctly?

Specificity

Specificity is another important metric in the evaluation of classification models, particularly in binary classification. It measures the **ability of a model to correctly identify negative instances**. Specificity is also known as the True Negative Rate. Formula is given by:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Specificity} = \frac{330}{330 + 60} = 0.846$$

F1-score is a harmonic mean of Precision and Recall.

F1-score is used to evaluate the overall performance of a classification model. It is the harmonic mean of precision and recall,

$$F1 - score = \frac{2}{\frac{1}{Recall} + \frac{1}{Precision}}$$

$$F1 - score = \frac{2}{\frac{1}{0.918} + \frac{1}{0.903}} = 0.9106$$

When to use Accuracy / Precision / Recall / F1-Score?

- ***Accuracy*** is used when the ***True Positives and True Negatives*** are more important. ***Accuracy*** is a better metric for ***Balanced Data***.
- Whenever **False Positive** is much more important use ***Precision***.
- Whenever **False Negative** is much more important use ***Recall***.
- ***F1-Score*** is used when the ***False Negatives and False Positives*** are important. ***F1-Score*** is a better metric for ***Imbalanced Data***.

Balanced & imbalanced dataset

- A balanced dataset has approximately **equal numbers of samples for each class**, leading to more reliable and unbiased model performance in machine learning.
- An imbalanced dataset features a significant **disparity in the count of samples between classes**, a common occurrence in real-world problems like **fraud detection** or **medical diagnosis**.

Confusion Matrix For Multi-class Classification

	Predicted Cat	Predicted Dog	Predicted Horse
Actual Cat	True Positive (TP)	False Negative (FN)	False Negative (FN)
Actual Dog	False Negative (FN)	True Positive (TP)	False Negative (FN)
Actual Horse	False Negative (FN)	False Negative (FN)	True Positive (TP)

	Predicted Cat	Predicted Dog	Predicted Horse
Actual Cat	8	1	1
Actual Dog	2	10	0
Actual Horse	0	2	8

In this scenario:

- **Cats:** 8 were correctly identified, 1 was misidentified as a dog, and 1 was misidentified as a horse.
- **Dogs:** 10 were correctly identified, 2 were misidentified as cats.
- **Horses:** 8 were correctly identified, 2 were misidentified as dogs.

	Predicted 0	Predicted 1
Actual 0	40	10
Actual 1	5	45

Model performance evaluation for regression models

Types of loss

In linear regression, there are four main types of loss, which are outlined in the following table.

Loss type	Definition	Equation
L₁ loss	The sum of the absolute values of the difference between the predicted values and the actual values.	$\sum actual\ value - predicted\ value $
Mean absolute error (MAE)	The average of L ₁ losses across a set of examples.	$\frac{1}{N} \sum actual\ value - predicted\ value $
L₂ loss	The sum of the squared difference between the predicted values and the actual values.	$\sum (actual\ value - predicted\ value)^2$
Mean squared error (MSE)	The average of L ₂ losses across a set of examples.	$\frac{1}{N} \sum (actual\ value - predicted\ value)^2$

Regression Performance Evaluation Metrics

- **Mean Absolute Error (MAE)**

- It is the average difference between the original and predicted values.
- It measures how far the predictions are from the actual output
- We want to minimize it.
- It doesn't give you an idea of the direction of the error since you are taking only the absolute values.
- It doesn't penalize large errors as much as compared to RMSE.

Mean Absolute Error (MAE)

- Mathematically, it is represented as:

$$\text{MAE} = \frac{1}{n} \sum_{j=1}^n |y_j - \hat{y}_j|$$

here,

- n is the number of observations
- y_j is the actual value for sample j
- $\hat{y}_j$ is the predicted value for sample j

Assignment

- Consider a regression problem where you are trying to predict the number of readers of the article, and let's say your test set has two observations **n= 2**.
- If actual number of readers, $y_j = [10, 5]$ and your model predicts, $\hat{y}_j = [8, 6]$ readers then find the following
 1. MAE
 2. RMSE
 3. R Squared/ coefficient of determination

Model performance evaluation for
clustering models

Clustering Performance Evaluation Metrics

- Clustering is the most common form of unsupervised learning.
- In clustering, you don't have any labels, just a set of features for observation.
- You aim to create clusters with **similar observations** clubbed **together** and dissimilar observations kept as far as possible.
- Clustering evaluation is not as trivial as counting the number of errors or the precision and recalls like supervised learning algorithms.

- Here, cluster validation is based on similarity or dissimilarity measures, such as the distance between cluster points.
- If the clustering algorithm separates dissimilar observations apart and similar observations together, then it has performed well.
- The two most popular evaluation metrics for clustering algorithms are the **Silhouette coefficient** and **Dunn's Index**, which you will explore next.

Silhouette Coefficient

$$s = \frac{b - a}{\max(a, b)}$$

- The Silhouette Coefficient is defined for each sample and is composed of two scores:
- a : The mean distance between a sample and all other points in the same cluster.
- b : The mean distance between a sample and all other points in the next nearest cluster.
- The Silhouette Coefficient for a set of samples is given as the mean of the Silhouette Coefficient for each sample.
- The score is between -1 for incorrect clustering and +1 for highly dense clustering.
- Scores around zero indicate overlapping clusters.
- The score is higher when clusters are dense and well separated, which relates to a standard cluster concept.

Dunn's Index

- Dunn's Index (DI) is another metric for clustering algorithm evaluation.
- Dunn's Index equals the **minimum inter-cluster distance divided by the maximum cluster size**.
- Large inter-cluster distances (better separation) and smaller cluster sizes (more compact clusters) lead to a higher DI value.
- A higher DI implies better clustering. It assumes that better clustering means that clusters are compact and well-separated from other clusters.



Thank you